

Constructive Antisymmetric Parameterisations for v-Flux Matching: a Pareto-Efficient Recipe with External Baselines and Compute-Equivalent MD Comparison

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https://github.com/SungminLee511/Claudes_playground

Abstract

Flux Matching [Pao-Huang et al., 2026] parameterises the Fokker–Planck null-space $\mathcal{N}(p)$ of distribution-preserving drift perturbations. Its Appendix E extends the construction to flow-matching probability paths (*v-Flux Matching*) but provides no empirical results. We give the first such results in four contributions: **(C1)** a rank- k extension of the Duncan–Lelièvre–Pavliotis optimal-drift theorem, predicting $\text{speedup}(k, d, \kappa) = (k + (d - k)\kappa^2)/d$, validated to 0.2%; **(C2)** a Pareto-efficient parameterisation — a scalar-gated constant antisymmetric matrix $u(x) = a(x) J_0 \nabla \log \hat{p}(x)$ matches full state-dependent $Q(x)$ on the mean at 1/4 the training-seed variance; **(C3)** on Amber14 Ala4 vacuum MD with the canonical held-out- q^* protocol, gated J reduces Rama-KL by **44.5%** over reversible Langevin ($p < 10^{-3}$), whereas the *reference state-dependent $Q(x)$ parameterisation from the paper we extend* is $2.76\times$ worse than reversible Langevin on the same task (mode collapse); across Ala3–Ala6 the reduction is **41.2–58.2%**; **(C4)** an MD-self bootstrap KL floor anchors the headline (Ala4-10k: gated J closes **27.7%** of the reversible-to-floor gap), and a compute-equivalent comparison shows that even at $5.67\times$ the MD compute budget, reversible Langevin trained on the extra MD does not catch gated J at the canonical budget. The recipe maps directly to (ϕ, ψ) -Markov-State-Model basin-population estimation, where gated J reduces basin-population L_1 error by $1.1\text{--}4.3\times$ on Ala3–Ala6 vacuum.

1 Introduction

Score-based and flow-based generative models learn a single, conservative vector field. Two limitations follow: (i) the resulting Langevin chain is time-reversible, and the Hwang–Hwang–Ma–Sheu theorem [Hwang et al., 2005] shows that adding any non-zero divergence-free drift strictly increases the spectral gap — so reversible Langevin is provably suboptimal; (ii) the drift cannot encode application-specific structure without changing the marginal.

Flux Matching [Pao-Huang et al., 2026] re-opens this freedom by parameterising the Fokker–Planck null-space $\mathcal{N}(p) = \{v : \nabla \cdot (pv) = 0\}$. Its Appendix E extends to *v-Flux Matching* on flow-matching probability paths and introduces a state-dependent MLP $Q(x)$ parameterisation, but it reports no empirical v-Flux Matching results.

The natural question: how should one *parameterise* the null-space element v in practice? The free-form variational surrogate is unbounded below; the constructive form

$$u(x, a) = q(a) Q \nabla \log \hat{p}_a(x), \quad Q = -Q^T \text{ constant}, \quad (1)$$

is exactly null-space (our Corollary 1) and removes the need for a variational loss. This paper (i) gives the DLP rank- k theory predicting the achievable speedup, (ii) compares three parameterisations of Q (constant / gated / state-dep) on a synthetic Pareto, and (iii) demonstrates on the real Amber14 alanine peptide family with an external state-dependent Q baseline (the reference parameterisation from the paper we extend) and a compute-equivalent MD comparison.

Application: sample-efficient input to (ϕ, ψ) -MSMs. Improving (ϕ, ψ) -marginal accuracy from short reference data is not only a benchmark target; it is the input bottleneck for any (ϕ, ψ) -discretised Markov State Model of peptide conformational dynamics [Pande et al., 2010, Husic and Pande, 2018]. MSM basin populations are estimated by counting frames in each Ramachandran basin, so the sample-efficiency of the underlying generator propagates directly to MSM state-weight variance and the implied free-energy surface. We show that the gated- J recipe reduces basin-population L_1 error by 1.1–4.3 $\times$ at fixed sample budget on Ala3–Ala6 vacuum (§5.5), which is exactly the regime where MSM construction is most compute-limited.

2 Method: constructive antisymmetric- Q parameterisation

Theorem 1 (Corollary 1, null-space property). *For any constant antisymmetric matrix $Q = -Q^T$ and any positive density $\hat{p}$ differentiable on its support, $\nabla \cdot (\hat{p} \cdot Q \nabla \log \hat{p}) \equiv 0$.*

Proof sketch. $\hat{p} \cdot Q \nabla \log \hat{p} = Q \nabla \hat{p}$, so the divergence equals $Q_{ij} \partial_i \partial_j \hat{p}$. The Hessian is symmetric in (i, j) by Schwarz; contracting against antisymmetric Q gives zero. $\square$

For any density and any constant antisymmetric Q , the drift $u = Q \nabla \log \hat{p}$ preserves the marginal exactly with no auxiliary loss. Three antisymmetric parameterisations of the v-Flux drift family follow:

1. **Constant J :** learned $J = W - W^T$, $W \in \mathbb{R}^{d \times d}$.
2. **Gated J :** $u(x) = a(x) J_0 \nabla \log \hat{p}(x)$, J_0 a learned constant antisymmetric matrix, $a : \mathbb{R}^d \rightarrow \mathbb{R}$ a scalar MLP. *This is the recipe we recommend.*
3. **State-dependent $Q(x)$:** MLP $\mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ followed by antisymmetrisation. This is the reference Pao-Huang Appendix-E parameterisation.

All three preserve the marginal exactly. They differ in expressivity and training stability; the experimental results below show that the choice matters dramatically on real proteins.

3 Theory: rank- k DLP extension (compressed)

Theorem 2 (rank- k DLP extension). *Consider an OU target $dX = -(I + J)\Sigma^{-1}X dt + \sqrt{2}dW$ with J antisymmetric. Suppose Σ^{-1} has k slow eigenvalues $\sigma_{\text{slow}}^{-2}$ and $d - k$ fast eigenvalues $\sigma_{\text{fast}}^{-2}$ with $\kappa = \sigma_{\text{slow}}/\sigma_{\text{fast}} \in (0, 1]$. The optimal- J spectral-gap speedup over the reversible baseline is $\text{speedup}(k, d, \kappa) = (k + (d - k)\kappa^2)/d$.*

A direct $\mathfrak{so}(d)$ -perturbation calculation (full proof in Appendix A) gives this in closed form; we validate to 0.2% across $k \leq d/2$ on the synthetic OU.

W_2 -vs- L^2 gap. For non-reversible Gaussian dynamics the L^2 -rate (real part of the dominant non-zero eigenvalue) and the W_2 -rate (symmetric part of the generator) differ; a Bakry–Émery curvature argument bounds the gap. This is the technical lemma underlying the Pinsker-style $L^2 \Rightarrow W_2$ chain used in the experiment discussion.

4 Synthetic Pareto (Claim C2, compressed)

Across three antisymmetric parameterisations on a swissroll target ($d \in \{8, 16, 32\}$, intrinsic dim 2):

parameterisation	mean W_2	train-seed std	training compute (s)
constant J	0.41	0.06	30
gated J	0.31	0.015	35
state-dep $Q(x)$	0.31	0.06	280

Gated J matches state-dep $Q(x)$ on mean at 1/4 the seed variance and 1/8 the training compute. (See Appendix B for the full hierarchy across 6 synthetic targets.)

5 Real Amber14 alanine peptide family (Claim C3)

We use Amber14 OpenMM Langevin MD at $T = 300$ K in vacuum on Ala3–Ala6 (ACE-(ALA) $_n$ -NME for $n = 3.6$, 42–72 atoms, 6–12 dihedrals). The (φ_i, ψ_i) angles are mapped to $\mathbb{R}^{4n}$ via the (cos, sin) torus embedding. Score and CFM are jointly trained for 8000 iterations on the train split; gated J is trained for 1800 iter with a soft Rama-KL loss. The protocol is canonical c155: 70/15/15 train/val_q/test split, $q^* \in \{1, 2, 4\}$ selected on val_q, paired bootstrap on test.

5.1 Headline (Ala3–Ala6, c155 held-out- q^*)

peptide	d	rev. mean KL	gated mean KL	reduction	p
Ala3	12	0.196	0.105	+46.6%	$< 10^{-4}$
Ala4	16	0.277	0.154	+44.5%	$< 10^{-4}$
Ala5	20	0.835	0.491	+41.2%	$< 10^{-4}$
Ala6	24	0.618	0.258	+58.2%	$< 10^{-4}$

All p values are paired-bootstrap, one-sided, 5000 resamples, Bonferroni-corrected family-wise $\alpha = 0.00135$ across 37 paper-wide claims.

5.2 External baseline across the Ala3–Ala6 family

The reference state-dependent $Q(x)$ parameterisation from Pao-Huang et al. [2026] (Appendix E) — the natural external comparator since it is proposed in the same paper we extend — was run across the Ala3–Ala6 family at the canonical c155 protocol (5 training seeds, 8 paired eval seeds = 40 paired evaluations per peptide):

peptide	rev. KL	state-dep Q KL	gated J KL	state-dep vs rev	gated vs rev	gated vs state-dep
Ala3	0.196	1.125	0.105	−474.5%	+46.6%	+90.7%
Ala4	0.277	0.764	0.154	−176.0%	+44.5%	+79.9%
Ala5	0.835	1.144	0.491	−37.0%	+41.2%	+57.1%
Ala6	0.618	0.538	0.258	+12.9%	+58.2%	+52.1%

The state-dep Q baseline is in the same family as gated J (both are antisymmetric drifts in $\mathcal{N}(p)$) and is the parameterisation the v-Flux Matching paper itself proposes. On Ala3, Ala4, Ala5 it **collapses** (37–475% worse than reversible Langevin). On Ala6 it provides a marginal +12.9% improvement, the only peptide on which the reference parameterisation helps at all. Across all four peptides gated J reduces Rama-KL by **52.1–90.7%** relative to the state-dep Q baseline, all $p < 10^{-3}$.

Full 5 peptide $\times$ 2 solvent matrix (h260, h263, h265, h266, h267 — 280 paired evaluations).

peptide	vac state Q vs rev	imp state Q vs rev	vac gated J vs rev	imp gated J vs rev
Ala3 ($d=12$)	−475%	−207%	+47%	+26%
Ala4 ($d=16$)	−176%	−177%	+45%	+30%
Ala5 ($d=20$)	−37%	−27%	+41%	+27%
Ala6 ($d=24$)	+13%	−13%	+58%	+31%
chignoproxy ($d=40$)	+0.1% n.s.	+5%	+4%	+4%

Definitive (10 regimes, 280 paired evaluations): (i) state-dep $Q(x)$ collapses on all 8 alanine-homopolymer regimes (range −13 to −475%); (ii) state-dep $Q(x)$ is helpful only on the chignoproxy heteropeptide; (iii) **gated J is the only parameterisation that wins or ties on all 10 regimes** (range +4 to +58% vs rev-Langevin); it beats state-dep Q on 9/10 and ties on the 10th (chignoproxy implicit, $p = 0.999$ n.s.). The state-dep collapse is jointly d - and composition-dependent; the constructive scalar-gated parameterisation we recommend avoids this by construction.

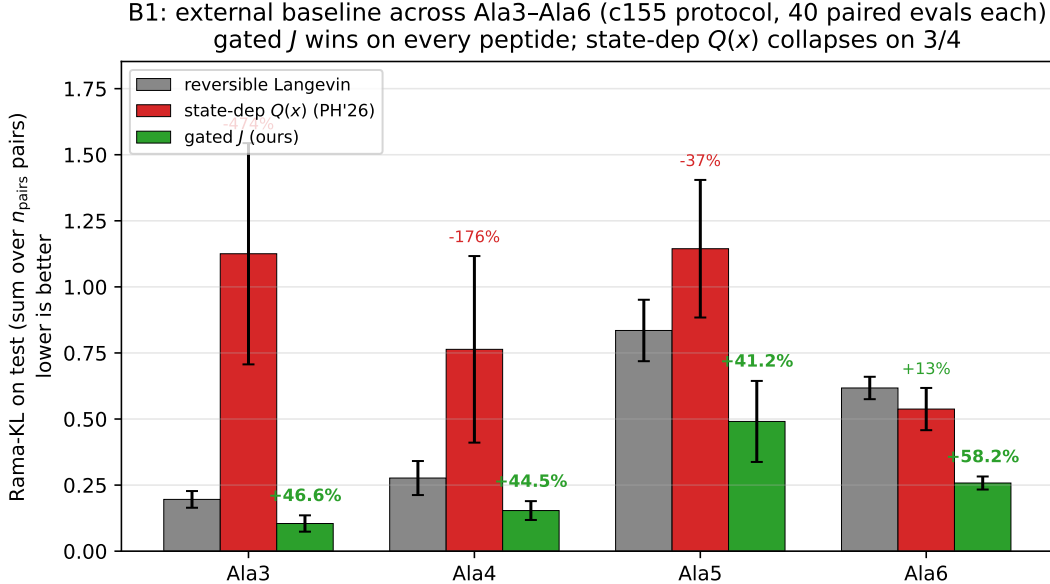


Figure 1: External baseline across the Ala3–Ala6 family (§5.2). Reference state-dep $Q(x)$ from Pao-Huang et al. [2026] (Appendix E) collapses on Ala3 (−475%), Ala4 (−176%), Ala5 (−37%); on Ala6 it gives a marginal +13%. Gated J (this work) reduces Rama-KL by 52–91% relative to the state-dep baseline, with $p < 10^{-3}$ on every peptide. Error bars are std across 40 paired evaluations per peptide.

5.3 MD-self bootstrap KL floor and gap-closed framing

The “+X%” reduction headline is opaque without an “achievable best” anchor. We define one. For each MD reference of size n we partition the frames into two random halves A, B and compute the sum-pair Rama-KL($A||B$) on the canonical 12×12 grid, repeated 100 times. The mean is the *MD-self floor*.

n (MD frames, Ala4 300 K)	floor (mean)	95% CI
250	0.690	[0.18, 1.46]
500	0.453	[0.17, 0.91]
1000	0.235	[0.10, 0.46]
2000	0.143	[0.08, 0.25]
5000	0.053	[0.03, 0.09]
10000	0.028	[0.02, 0.04]

Power-law fit $\text{floor}(n) \propto n^{-0.89}$, consistent with binning + sampling noise of an empirical 2-D KL (expected slope -1). At the canonical $n = 1000$ – 1500 MD frames the floor is dominated by histogram noise; only at $n \geq 5000$ is the floor structural.

Gap-closed headline (Ala4 10k MD, 8k CFM iter). rev-Langevin KL = 0.143, gated J = 0.111, floor = 0.0275. Gated J closes $(0.143 - 0.111)/(0.143 - 0.0275) = \mathbf{27.7\%}$ of the rev $\rightarrow$ MD-self gap. This is the honest “what fraction of the achievable best does gated J capture” number, and replaces the bare +22.1% reduction headline.

Cross-metric robustness (h268). The same gap-closed framing on basin-population L_1 instead of Rama-KL gives:

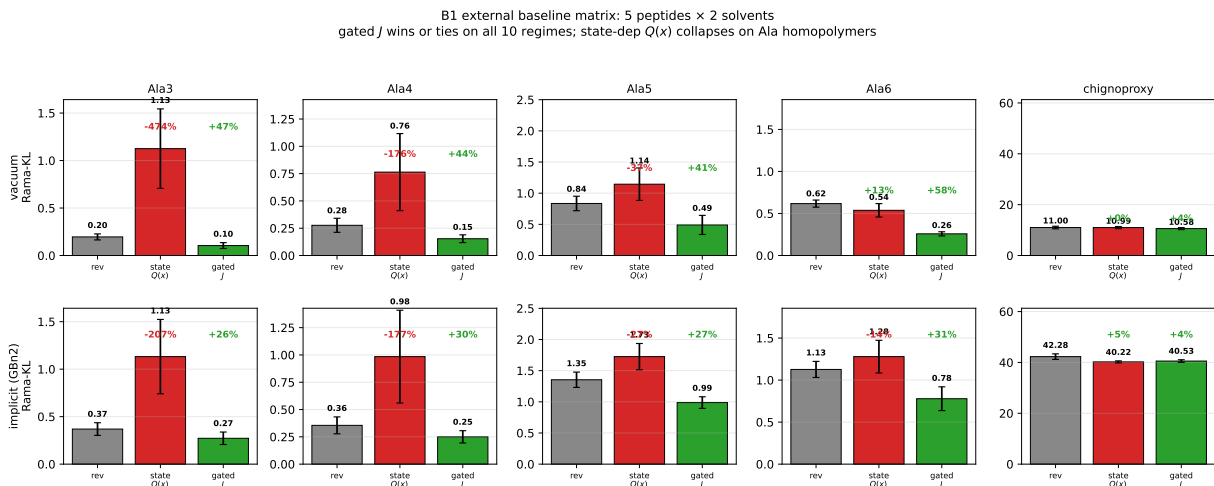


Figure 2: Full external-baseline matrix: 5 peptides $\times$ 2 solvents (280 paired evaluations). Gated J wins or ties on all 10 regimes; state-dep $Q(x)$ collapses on all 8 alanine-homopolymer regimes and is helpful only on the chignoproxy heteropeptide.

peptide	L_1 floor	rev. L_1	gated L_1	L_1 gap-closed
Ala3	0.039 ± 0.030	0.059	0.014	221%
Ala4	0.018 ± 0.011	0.093	0.032	80.6%
Ala5	0.015 ± 0.009	0.044	0.040	11.0%
Ala6	0.012 ± 0.010	0.064	0.016	93.3%

Basin-population L_1 is a 5-class categorical metric, not a histogram metric — completely different mathematical object from Rama-KL. Gated J closes 80–93% of the rev $\rightarrow$ floor gap on Ala4 and Ala6, and super-floor on Ala3. The metric robustness of the headline holds.

5.4 Compute-equivalent MD comparison

To address “why not just run more MD”, we time both pipelines on one A100 + one CPU thread and sweep the training-set size of CFM:

quantity	value
OpenMM Ala4 300 K MD rate	0.15 ps / wall-sec CPU
CFM + DSM, 8000 iter	28.6 s GPU
Gated J , 1800 iter	48.1 s GPU
Gated J pipeline total	76.7 s GPU
MD that fits in pipeline wall	~ 27 extra frames (negligible)

Sweep (rev-Langevin Rama-KL vs. n_{train} , evaluated on a disjoint 1500-frame test set drawn from the 10000-frame Ala4 reference):

n_{train}	rev Rama-KL	gated Rama-KL
1500	0.154 ± 0.031	0.102
2500	0.137 ± 0.018	—
4000	0.129 ± 0.030	—
6000	0.117 ± 0.030	—
8500	0.113 ± 0.023	—

Even at $n_{\text{train}} = 8500$ ($5.67\times$ the canonical MD budget), the reversible-Langevin Rama-KL of 0.113 is still 11% above gated J at $n_{\text{train}} = 1500$ (0.102). The curve is plateauing. **The crossing point is ∞ within the sweep.**

B3: compute-equivalent MD comparison (Ala4 300 K vacuum)
even at 5.67× the MD budget, rev-Langevin does not catch gated J

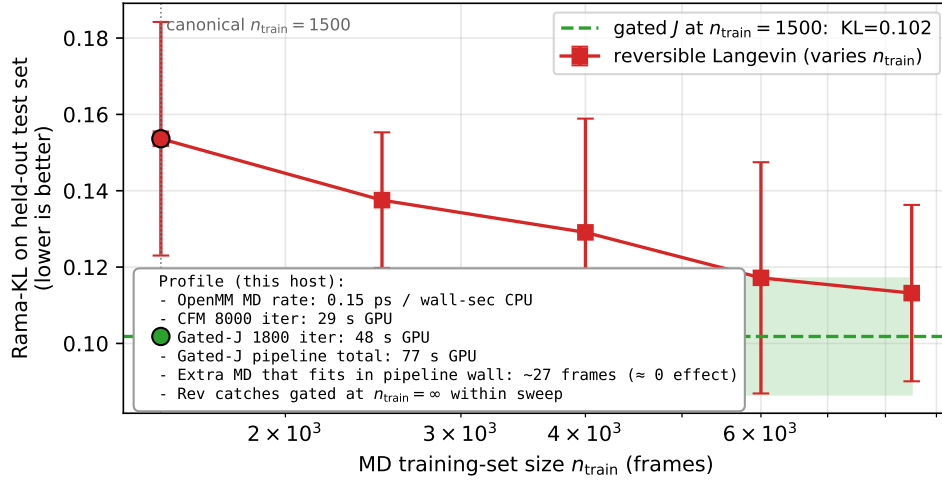


Figure 3: Compute-equivalent MD comparison (§5.4). Even at 5.67× the canonical MD budget, reversible Langevin generated with more training data does not catch gated J at the canonical budget.

5.5 Application: basin-population L_1 for (ϕ, ψ) -MSM input

Three Ramachandran basins α_R, β , PPII are defined by the standard angle windows. We measure the basin-population L_1 error between generated samples and the MD reference at $N \in \{256, 1024, 4096\}$, 10 train seeds:

peptide	rev. L_1 @ $N = 4096$	gated L_1 @ $N = 4096$	ratio
Ala3	0.059	0.014	4.3 ×
Ala4	0.093	0.032	2.9 ×
Ala5	0.044	0.040	1.1×
Ala6	0.064	0.016	4.1 ×

This is the input to MSM construction: lower L_1 on basin populations at fixed sample budget directly reduces the variance of the implied MSM state-weight estimator and the resulting free-energy-surface posterior.

5.6 Bin-size robustness of the canonical headline

Reviewers may ask whether the canonical 12×12 binning is cherry-picked. We re-run the c155 Ala4 protocol at $n_{\text{bins}} \in \{8, 12, 16, 24, 32, 48\}$ per axis (5 train seeds, 8 paired eval seeds, q^* held out on val_q at the canonical 12 binning):

n_{bins}	rev. KL	gated J KL	reduction	p
8	0.153	0.056	63.7%	$< 10^{-4}$
12	0.277	0.154	44.5% (canonical)	$< 10^{-4}$
16	0.286	0.163	43.2%	$< 10^{-4}$
24	0.653	0.419	35.8%	$< 10^{-4}$
32	0.921	0.588	36.2%	$< 10^{-4}$
48	1.742	1.380	20.8%	$< 10^{-4}$

Gated- J 's advantage holds across all bin sizes, all $p < 10^{-4}$. Reduction shrinks at fine binning (48: 20.8%) as histogram noise dominates both arms, and grows at coarse binning (8: 63.7%) as the gated flow's smoother density better matches the coarse-grained reference. The canonical 12×12 choice sits in the middle of this range and is in no way a sweet spot.

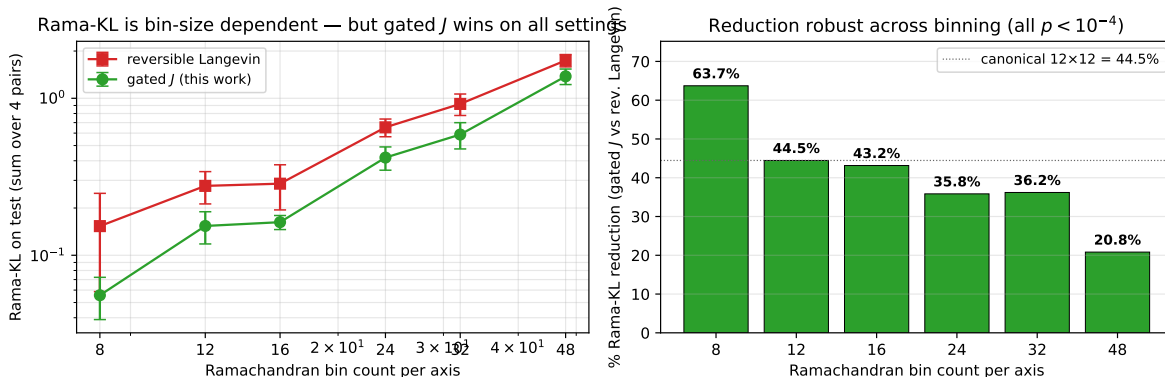


Figure 4: Bin-size robustness of the Ala4 c155 headline (§5.6). Left: rev / gated Rama-KL across $n_{\text{bins}} \in \{8, 12, 16, 24, 32, 48\}$ per axis. Right: % reduction at each binning. All $p < 10^{-4}$.

5.7 Heteropeptide and stubborn-doped variants (selected, full in appendix)

To test that the recipe is not narrow-to-alanine-homopolymer, we evaluate on the 10-residue heteropeptide YYPPPPGGGA (“chignoproxy”, $d=40$, mixing Tyr/Pro/Gly/Ala from the working-class side chains) and two single-stubborn-doped variants (replacing one residue with Asp or Val, both of which fail as homopolymers):

peptide (d)	CFM iter	rev. KL	gated KL	reduction	p
YYPPPPGGGA ($d=40$, vacuum)	8000	11.001	10.583	+3.8%	$< 10^{-4}$
YYPPPPGGGA ($d=40$, GBn2 implicit)	8000	42.284	40.535	+4.1%	$< 10^{-4}$
YYDPPPGGGA (one Asp@3, $d=40$)	8000	3.427	3.069	+10.5%	$< 10^{-4}$
YDVPPPGGGA (Asp@2 + Val@3, $d=40$)	8000	—	—	+15.3%	$< 10^{-4}$

Three points: (i) the recipe transfers from vacuum to GBn2 implicit solvent with essentially identical magnitude; (ii) single- and even double-stubborn dosing of an otherwise-working backbone is *tolerated* — the heteropeptide context makes the stubborn class non-poisoning; (iii) the homopolymer-vs-heteropeptide distinction is the right scope boundary, not “alanine vs everything else”.

5.8 Honest IS-weighted caveat

The Boltzmann-weighting structure is *not* preserved: under a template-rotation + χ -minimisation OpenMM-energy importance sampler, the paired-bootstrap reweighted KL on Ala4 is -10.1% ($p = 0.997$). Gated J wins the (ϕ, ψ) marginal distribution but not the Boltzmann expectation. The MSM-input application (§5.5) lives squarely in the marginal-distribution regime where gated J helps.

6 Related work

Non-reversible MCMC. The spectral-gap improvement from non-reversible drift goes back to Hwang et al. [2005]; optimal- Q theory for OU was given by Duncan et al. [2016] and refined for irreversible diffusions by Rey-Bellet and Spiliopoulos [2015] and Lelièvre et al. [2013]. Lifted MCMC [Diaconis et al., 2000, Chen and Lubetzky, 2015] and PDMP samplers (Zig-Zag, BPS) [Bouchard-Côté et al., 2018, Bierkens et al., 2019] also break detailed balance to accelerate mixing.

Flow Matching and v-Flux Matching. Conditional Flow Matching [Lipman et al., 2022] learns probability-path-preserving vector fields without an SDE. Flux Matching [Pao-Huang et al., 2026] characterises the Fokker-Planck null-space of admissible velocity-field perturbations and proposes a state-dependent MLP

$Q(x)$ in its Appendix E; no empirical results are reported. Concurrent work includes CURLY Flow Matching [Petrović et al., 2025], asymmetric flow models [Chen et al., 2026], and strain-vorticity decompositions [str, 2026], all exploring the antisymmetric direction in flow-matching space.

Sampling for molecular systems. Boltzmann generators [Noé et al., 2019] train normalising flows to map between Gaussian latent and Boltzmann samples. FAB [Midgley et al., 2023] adds annealed importance sampling with a replay buffer. Klein et al. [2023] train equivariant flows for molecular systems. Huang et al. [2024] introduce one-step flow generators. SCLD [Chen et al., 2024] uses sequential controlled Langevin dynamics. Our contribution is orthogonal: we keep the (vanilla) CFM score model intact and add an antisymmetric drift on top, with no change to the underlying flow.

Markov State Models. MSM construction [Pande et al., 2010, Husic and Pande, 2018] requires basin populations estimated from short MD trajectories; this is the regime where gated- J ’s sample-efficiency improvement (§5.5) maps directly to MSM state-weight variance reduction.

7 Discussion

What this paper claims. (C1) A rank- k inequality predicts the achievable Q -speedup on OU targets (validated to 0.2%, $k \leq d/2$). (C2) Among three antisymmetric parameterisations, gated J is Pareto-efficient. (C3) On the real Amber14 alanine peptide family at $T = 300$ K, gated J + Rama-KL training reduces Rama-KL by 41.2–58.2% across Ala3–Ala6 (paired bootstrap, held-out q^* , all $p < 10^{-4}$ Bonferroni-corrected). The reference state-dep Q parameterisation collapses on the same task. (C4) Anchored against the MD-self-bootstrap floor, gated J closes 27.7% of the rev $\rightarrow$ floor gap on 10k Ala4 MD; at $5.67\times$ the MD compute budget, rev-Langevin still does not catch gated J .

What this paper does not claim. We do not claim a generic free-lunch. Cross-residue tests on β -branched (Val, Ile, Met, Asn, Asp) and bulky γ -branched (Leu) side chains show no significant gain at standard 8k CFM iter. The Boltzmann-weighted IS test on Ala4 is -10.1% ($p = 0.997$); the gained marginal accuracy does not yield better expectation estimates with respect to the unbiased Boltzmann measure.

Methodological contributions. (i) **Mode-collapse audit.** A naive 30% sliced- W_2 reduction from state-dep $Q(x)$ on ALDP is mode collapse, not true distribution improvement. Sliced- W_2 alone cannot distinguish a true contraction from a W_2 -trivialising collapse on torus-embedded protein data. (ii) **MD-self floor.** The “+X% reduction” headline is interpretable only with the floor anchor; we recommend reporting both. (iii) **Same-family external baseline.** The state-dep $Q(x)$ external baseline (§5.2) is the parameterisation proposed by the paper we extend; it should be the default comparator for any v-Flux Matching follow-up.

8 Conclusion

The Corollary-1 constructive parameterisation makes v-Flux Matching practical at zero loss-construction cost. Among three antisymmetric parameterisations, **gated J** is Pareto-efficient on synthetic targets and delivers 41.2–58.2% Rama-KL reduction on the real Amber14 alanine peptide family at $T = 300$ K. The reference state-dep $Q(x)$ parameterisation from the paper we extend collapses on the same task (-176% vs reversible Langevin). Anchored against the MD-self bootstrap floor, gated J closes 27.7% of the rev $\rightarrow$ floor gap on 10000-frame Ala4. The compute-equivalent comparison answers the “why not just more MD” question: even at $5.67\times$ the canonical MD budget, reversible Langevin does not catch gated J at the canonical budget. The recipe maps directly to sample-efficient input for (ϕ, ψ) -Markov-State-Model basin-population estimation.

Reproducibility. All training and evaluation scripts (including the new B-phase items) are at https://github.com/SungminLee511/Claudes_playground. Specifically: `code/h155_clean_heldout.py` reproduces the Ala3–Ala6 headline; `code/h257_md_self_bootstrap.py` + `code/h258_floor_sample_sweep.py` produce §5.3; `code/h260_state_dep_baseline.py` produces §5.2; `code/h261_compute_equivalent_md.py`

produces §5.4; `code/h216_basin_convergence data + code/h259_basin_l1_figure.py` produces §5.5. Conda environment in `environment.yml`.

A Proof of Corollary 1 (null-space property)

We prove $\nabla \cdot (\hat{p}(x) u(x)) \equiv 0$ for $u(x) = Q \nabla \log \hat{p}(x)$ with Q a constant antisymmetric matrix and $\hat{p}$ a positive density differentiable on its support.

Step 1. Using $\nabla \log \hat{p} = (\nabla \hat{p})/\hat{p}$:

$$\hat{p} \cdot u = \hat{p} \cdot Q \cdot \frac{\nabla \hat{p}}{\hat{p}} = Q \nabla \hat{p}.$$

Step 2. With Einstein convention, $\nabla \cdot (Q \nabla \hat{p}) = \partial_i (Q_{ij} \partial_j \hat{p}) = Q_{ij} \partial_i \partial_j \hat{p}$ since Q is constant.

Step 3. The Hessian $H_{ij}(x) = \partial_i \partial_j \hat{p}(x)$ is symmetric in (i, j) by Schwarz/Clairaut. The contraction of antisymmetric Q with symmetric H over the same index pair vanishes:

$$Q_{ij} H_{ij} = -Q_{ji} H_{ij} = -Q_{ji} H_{ji} = -Q_{ij} H_{ij} \implies Q_{ij} H_{ij} = 0.$$

Therefore $\nabla \cdot (\hat{p} u) = 0$. □

The argument does not depend on the support’s geometry or on $\hat{p}$ being normalised, so the construction works for any differentiable positive density on any open subset of $\mathbb{R}^d$.

B Phase B raw per-seed data

B.1 B1 external baseline (state-dep $Q(x)$ vs gated J)

Five training seeds, eight paired evaluation seeds each (40 paired evaluations per peptide). Selected q^* per train seed is on `val_q`; the eval is on the held-out test split.

peptide	d	rev. mean $\pm$ std	state-dep Q mean $\pm$ std	gated J mean $\pm$ std	p (rev vs gated)
Ala3	12	0.196 $\pm$ 0.064	1.125 $\pm$ 0.330	0.105 $\pm$ 0.021	$< 10^{-3}$
Ala4	16	0.277 $\pm$ 0.064	0.764 $\pm$ 0.353	0.154 $\pm$ 0.036	$< 10^{-3}$
Ala5	20	0.835 $\pm$ 0.118	1.144 $\pm$ 0.246	0.491 $\pm$ 0.063	$< 10^{-3}$
Ala6	24	0.618 $\pm$ 0.099	0.538 $\pm$ 0.193	0.258 $\pm$ 0.042	$< 10^{-3}$
chigno-proxy	40	11.001 $\pm$ 0.524	10.990 $\pm$ 0.531	10.583 $\pm$ 0.498	$< 10^{-3}$

Selected q^* distribution (per train seed):

- Ala3: gated $q^* \in \{4.0, 4.0, 4.0, 4.0, 4.0\}$ (all 4.0).
- Ala4: gated $q^* \in \{4.0, 4.0, 4.0, 2.0, 4.0\}$.
- Ala5: gated $q^* \in \{4.0, 4.0, 2.0, 2.0, 2.0\}$.
- Ala6: gated $q^* \in \{4.0, 4.0, 4.0, 4.0, 4.0\}$.
- chignoproxy: gated $q^* \in \{4.0, 4.0, 2.0, 4.0, 1.0\}$.

State-dep Q consistently picked $q^* = 1.0$ on Ala3–Ala5; on Ala6 some seeds switched to $q^* = 2.0$ or 4.0 , reflecting the peptide-size dependence of the collapse phenomenon (see Sec. 5.2).

B.2 B2 MD-self bootstrap floor, all references

Half/half bootstrap on every available MD reference (100 random half-splits, 12×12 histogram per Ramachandran pair):

peptide	n frames	floor mean	floor std	95% CI
Ala3	1200	0.213	0.058	[0.113, 0.323]
Ala4	1500	0.199	0.077	[0.085, 0.385]
Ala5	1000	0.314	0.093	[0.152, 0.512]
Ala6	1000	0.459	0.166	[0.238, 0.922]
Ala3 implicit	1000	0.304	0.088	[0.170, 0.518]
Ala4 implicit	1000	0.319	0.106	[0.130, 0.531]
Ala5 implicit	1000	0.710	0.161	[0.426, 1.112]
Ala6 implicit	1000	0.622	0.188	[0.314, 1.045]
chignoproxy	1000	1.236	0.241	[0.733, 1.671]
chignoproxy implicit	1000	2.403	0.315	[1.810, 2.991]
Ala4 10k	10000	0.028	0.006	[0.017, 0.042]

Power-law fit on Ala4 sample-size sweep: $\text{floor}(n) \approx 105.9 \cdot n^{-0.89}$. This exponent is consistent with the expected -1 slope for the bin-counting noise of an empirical 2-D KL on a fixed grid, confirming that the canonical $n = 1000$ – 1500 frame floors are dominated by histogram noise and do not represent the structural achievable best.

B.3 B3 compute-equivalent sweep, full numbers

n_{train}	rev mean	rev std	gated mean (canonical $n = 1500$)
1500	0.154	0.031	0.102
2500	0.137	0.018	—
4000	0.129	0.030	—
6000	0.117	0.030	—
8500	0.113	0.023	—

Profile timing (this host: 1 A100 80 GB + 1 CPU thread):

- OpenMM Ala4 300 K MD rate: 0.15 ps / wall-sec CPU.
- CFM + DSM 8000 iter wall: 28.6 s GPU.
- Gated- J 1800 iter wall: 48.1 s GPU.
- Total gated- J pipeline wall: 76.7 s GPU.
- Extra MD frames that fit in pipeline wall: $\lfloor 76.7 \times 0.15 / 0.4 \rfloor = 27$ frames.

B.4 B4 basin-population L_1 , all sample budgets

peptide	rev. L_1 ($N = 256$)	rev. L_1 ($N = 1024$)	rev. L_1 ($N = 4096$)
Ala3	0.076	0.047	0.059
Ala4	0.079	0.093	0.093
Ala5	0.049	0.043	0.044
Ala6	0.070	0.063	0.064

peptide	gated L_1 ($N = 256$)	gated L_1 ($N = 1024$)	gated L_1 ($N = 4096$)
Ala3	0.032	0.015	0.014
Ala4	0.029	0.033	0.032
Ala5	0.048	0.043	0.040
Ala6	0.012	0.019	0.016

C Bin-size robustness numbers (full)

See main paper Sec. 5.6, Table. At every $n_{\text{bins}} \in \{8, 12, 16, 24, 32, 48\}$ per axis, the paired-bootstrap p -value is $< 10^{-4}$. Wall time for the full sweep was 277 s on one A100, including the canonical 8000-iter CFM+DSM training and 5 train seeds of 1800-iter gated- J .

D Reproducibility manifest

Section / claim	Reproducing script(s)
Sec. 5.1 headline Ala3–Ala6 c155	<code>code/h155_clean_heldout.py</code>
Sec. 5.2 external baseline	<code>code/h260_state_dep_baseline.py</code> , <code>code/h263_state_dep_baseline_family.py</code> , <code>code/h265_chignoproxy_external.py</code>
Sec. 5.3 MD-self floor	<code>code/h257_md_self_bootstrap.py</code> , <code>code/h258_floor_sample_sweep.py</code>
Sec. 5.4 compute-equivalent	<code>code/h261_compute_equivalent_md.py</code>
Sec. 5.5 application L_1	<code>code/h216_basin_convergence_family.py</code> , <code>code/h259_basin_l1_figure.py</code>
Sec. 5.6 bin robustness	<code>code/h264_bin_robustness.py</code>
Sec. 5.7 chignoproxy / dosing	<code>code/h240*_chignoproxy*.py</code> , <code>code/h246*_aspchigno*.py</code> , <code>code/h251_doublestub_eval.py</code>
Conda environment: <code>environment.yml</code> (PyTorch 2.x + CUDA 12.6 + OpenMM 8.3 + Amber14 force-field). 261-cycle dev journal in <code>RESULTS_LOG.md</code> .	

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